The Nine Guidelines and Their Rivals

Session VI · 12 October 2026 — the five-step pipeline, reverse causation, the OxyContin case

Class notes · an automated summary of the live session, structured for study. Where these notes and the assigned papers disagree, the papers win.

Concept

- Causal inference is the process of judging whether an observed association between an exposure (X) and an outcome (Y) reflects a true cause-and-effect relationship
- This judgment is not absolute truth — it is a conclusion based on available evidence, assumptions, and the analytical tools accessible at a given point in history
- Hill's guidelines (nine criteria developed by Hill, as presented in the Gordis Epidemiology textbook) are frameworks for thinking through causality — they are not a checklist where all nine must be satisfied
- Four core rivals (alternative explanations) can undermine any causal claim: random chance, bias (systematic error), confounding (a third factor), and reverse causation

What I understood

The Five-Step Pipeline for Evaluating Causality

1. Define the exposure (X): Specify what it is and how it is measured (e.g., infection defined by elevated temperature, leukocytes, C-reactive protein, or a specific diagnostic test)

2. Define the outcome (Y): Specify and measure it in the same rigorous way

3. Establish the association: Connect X and Y — this does not yet imply causation, only a statistical relationship

4. Name the rivals: Identify what could alternatively explain the association (chance, bias, confounding, reverse causation)

5. Decide and state what would undo it: Articulate what evidence or alternate explanation would overturn your conclusion — it is acceptable to remain undecided if evidence is insufficient

Hill's Nine Guidelines and How Each Can Be Undermined

- Temporal relationship — X must precede Y; this is the only criterion that can single-handedly rule out causality
  - Destroyed exclusively by reverse causation
  - Example: COVID-19 vaccine blamed for causing infection because symptoms appeared after vaccination — but the patient was likely exposed to the virus before vaccination, and the immune response takes at least 7 days to develop
- Strength of association — expressed as how many times more (or less) often the outcome occurs in the exposed group
  - Smoking increases lung cancer risk ~20× compared to non-smokers
  - Can be distorted by chance, bias in how groups were selected or outcomes counted, or confounding factors
- Dose-response relationship — more exposure leads to more outcome (e.g., heavier smoking → higher cancer risk); closely linked to cessation of exposure
  - If exposure stops and outcome frequency decreases, this supports causality
  - Can be disrupted by chance, bias, or confounding (e.g., a cultural tradition of consuming a particular substance in a region)

  - Counter-example: genetic predisposition to lung cancer means the outcome may occur regardless of whether smoking stops — some causes are independent of exposure cessation
- Replication — the same result reproduced across multiple studies
  - Critical rule: Replication only counts if the study designs differ; reproducing results using the same design repeatedly does not validate causality — it may simply replicate the same systematic error
- Biological plausibility — a credible biological mechanism explains the connection
  - Considered a weak criterion because humans can construct plausible biological explanations for almost anything
  - Its strength: it cannot be undermined by chance, bias, confounding, or reverse causation — making it somewhat independent, even if weak
- Cessation of exposure — outcome decreases when exposure is removed (supports causality)
  - Example: removing industrial toxins from a water supply (as in the Erin Brockovich case) reduces disease frequency
- Specificity — one cause produces one specific effect
  - Most relevant for infectious diseases (e.g., rabies virus causes only rabies)
  - Less applicable to multifactorial conditions like heart disease or lung cancer, which have multiple contributing causes — these represent non-sufficient causal relationships
- Analogy / Other knowledge — alternate explanations from existing knowledge
  - If an alternate explanation exists and cannot be ruled out, it must be incorporated into the study design and reported separately

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Reverse Causation — A Key Rival

- Occurs when the assumed direction of causality is inverted: you believe X causes Y, but in reality Y causes X
- COVID-19 vaccine example: Vaccine appeared to "cause" COVID-19 because symptoms emerged post-vaccination; the actual sequence was pre-vaccination viral exposure → vaccination → symptom onset (virus incubation preceded immune response)
- Cancer and weight loss example: Thin body composition might appear to "cause" cancer, but cancer-related metabolic changes actually cause the weight loss — the outcome is driving the apparent exposure
- Particularly prevalent in sociological and media research, where temporal relationships are harder to establish

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The OxyContin / Opioid Epidemic Case Study

- Setup: Konstantin's case examined whether pharmaceutical marketing caused the opioid overdose epidemic in the United States
- Conclusion reached: A causal relationship exists but the evidence is not strong, due to several interfering factors
- Key problems with establishing causality:
  - Incomplete marketing data: Much of the marketing information was proprietary and inaccessible
  - Reverse causation in marketing data: OxyContin was launched in 1996; prescribing peaked ~2010; marketing data used in the study only emerged ~2015 — meaning the marketing data appeared after the prescribing peak, suggesting it may reflect already-established prescribing practices rather than causing them
  - Cessation failure: When some marketing campaigns were legally banned, the epidemic did not end — indicating other factors were sustaining it independent of marketing

  - Confounding (practice as a third factor): The opioid crisis was largely a US-specific phenomenon because aggressive opioid prescribing was an established cultural/medical practice there, not universal
- Doctor reward data (a specific marketing proxy):
  - US law requires pharmaceutical companies to publish payments made to doctors who prescribe their products
  - Doctors who received any opioid-related payment in 2014 had ~10% more opioid prescriptions than those who did not
  - Meals paid for by pharma companies were also associated with higher OxyContin prescriptions
  - However, this reward data was itself interpreted as a reflection of the decade-long prescribing trend rather than a driver — another instance of reverse causation
- Historical precedent contributing to the crisis: A 1980 letter published in the New England Journal of Medicine (Porter & Jick) reported that hospital patients receiving opioids for chronic pain rarely developed addiction; this single letter was widely cited and used to justify expanding OxyContin prescriptions to outpatients — a foundational misapplication of limited evidence

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When You Cannot Decide

- At certain points in scientific history, available tools and study designs make it impossible to rule out rivals or definitively establish causality — this is a valid and important conclusion to state
- Example: Hormone replacement therapy (HRT) for menopausal women remains debated; evidence supports protective effects for specific hormone combinations and patient groups, but oral estrogen alone is associated with hormone-related cancers — the causal picture is still unresolved

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What I didn't fully get

Reverse Causation — Why It's Easy to Miss

- The intuition of "what came first" seems simple, but in practice the recorded order of events often differs from the actual biological or causal order
- In the COVID-19/vaccine example, the visible sequence (vaccination → symptoms) masked the actual sequence (viral exposure → vaccination → symptom onset)
- In epidemiological studies, the time dimension must be explicitly accounted for in the study design — it is not enough to assume temporal order from observation alone

Why Replication Across Same Designs Doesn't Count

- If ten case-control studies all find the same result, this is not true replication — they may all share the same design-level systematic error (bias built into how cases and controls are selected or measured)
- True replication requires the same finding to emerge from different study types (e.g., a randomized controlled trial AND an observational cohort study AND a case-control study all pointing the same direction)

Biological Plausibility as a Weak Criterion

- It may seem like having a biological explanation should be strong evidence — but the reason it is considered weak is that researchers can almost always construct a plausible mechanism post-hoc
- Its independence from the four main rivals (chance, bias, confounding, reverse causation) is what gives it any value at all — it cannot be "destroyed" by those rivals, but it also cannot confirm causality on its own

The OxyContin Timeline Complexity

- The 74–97% figure mentioned briefly at the end of the session (regarding the proportion of the association that does not run through prescribing) was not fully explained due to time constraints — this will be revisited in a future lesson

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Might show up on the exam

- The four rivals to any causal claim (must know all four and be able to apply them):
  - Random chance
  - Bias / systematic error (design-level)
  - Confounding (third factor associated with both exposure and outcome)
  - Reverse causation
- Temporal relationship is the only Hill criterion that alone can definitively rule out causality — if X does not precede Y, stop; no further causal argument is possible
- Replication validity rule: Same result across different designs = replication; same result across same designs = repeated systematic error, not evidence
- Reverse causation definition: You conclude X → Y, but the true direction is Y → X (or Y precedes X in a way that makes X appear causal)
- Biological plausibility: Weakest criterion; cannot be undermined by rivals but also cannot confirm causality — know why it is weak
- Specificity: Most relevant for infectious diseases; multifactorial diseases (heart disease, lung cancer) are inherently non-specific and require compound/constellation causal models
- Dose-response + cessation are linked: If more exposure → more outcome, AND removing exposure → less outcome, this is strong (but not conclusive) support for causality
- Confounding vs. reverse causation: Know the difference — confounding is a third factor associated with both X and Y; reverse causation is the direction of X and Y being inverted
- Undecidability is a valid scientific position: If rivals cannot be ruled out with available tools, stating that causality is undecidable at this point in time is scientifically appropriate
- Essay topic proposal is required for access to the final essay submission — a 3–4 sentence paragraph explaining topic choice must be submitted via private message by end of today's session

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Check before you lean on it

- The nine guidelines the course uses are Gordis's list (Gordis Epidemiology, ch. 14). Hill's own 1965 list differs in composition — the trainer follows Gordis, and so do these notes.
- The famous letter behind outpatient opioid expansion is Porter & Jick, NEJM 1980 — one paragraph, inpatients only (the transcript had placed it in Nature, 1960).
- "74–97% of the association does not run through prescribing" matches the mediation figures in the county-level study (3–26% mediated, depending on the marketing measure); the full walkthrough is in the course trainer's case pack.